

Example Approach Sequence Analysis

Step 1: Collect information about the sequence (2 to 5 days)

- Experimental results
 - How was it sequenced
 - Organism of origin
 - Expression organism
 - Missing parts
 - ...
- Secondary databases like InterPro

Step 2: Database Search (0.5 to 2 days)

- Run blast with default settings
- If the results aren't good enough → adjust the search:
 - Use other variants (PSI, PHI, DELTA)
 - Use different databases (First step)
 - Switch the substitution matrix

Step 3: Creating an MSA (~1 day)

- Use only "good" results, don't forget to add your sequence
- Analyze the MSA by taking into account the information from step 1

Step 4: Creating some pairwise-Alignments (~1 day)

- Use your sequence and 2-5 of the best sequences from Step 2 & Step 3
- The most important parts are the conserved regions / domains / motifs